DO NOW — RECALL

Lesson 155 · PageRank: Iterating a Link Graph

Name: _____ Class: _____ Date: _____

1. What does the inverted index map?

2. What is the role of a router?

3. What does a cache hit mean?

4. When can binary search be used?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Search terms to document IDs.

2. Model answer

Forward packets between IP networks.

3. Model answer

Required data already present in cache.

4. Model answer

On sorted data.

By the end, you can…

• Web pages are vertices; directed hyperlinks are edges.

• Each page shares rank across its outgoing links.

• PRnew=(1-d)/N+d×sum(incoming PR/outdegree).

Vocabulary — know these terms

PageRank — Link-based iterative importance algorithm.

directed graph — Nodes joined by one-way edges.

outdegree — Number of a page's outgoing hyperlinks.

damping — Factor balancing link following and random jumping.

NEW LEARNING

Read, annotate and explain

Build graph

Web pages are vertices; directed hyperlinks are edges.

Distribute rank

Each page shares rank across its outgoing links.

Apply damping

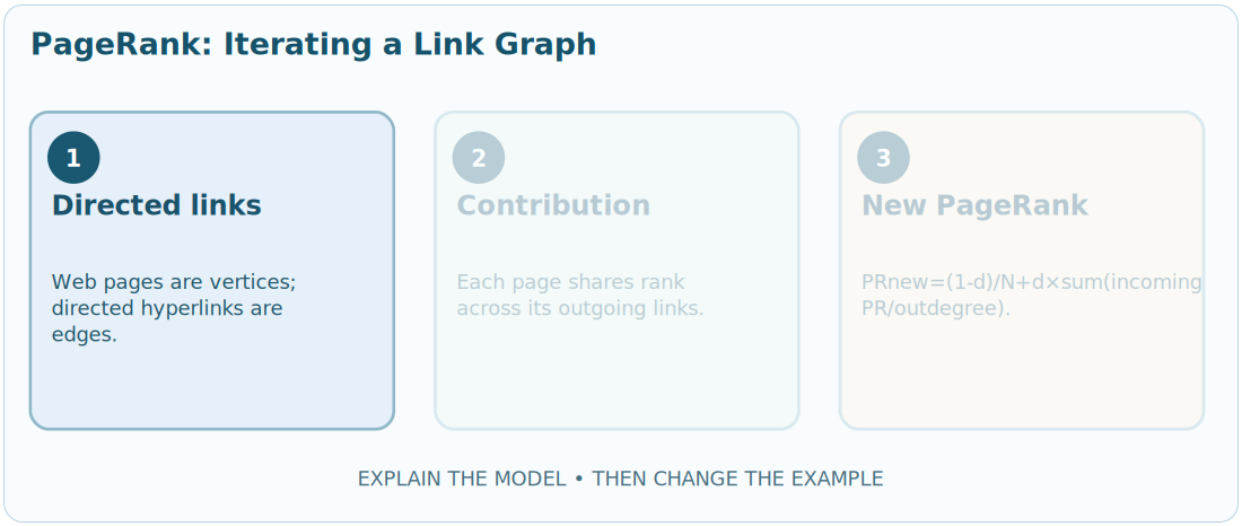
PRnew=(1-d)/N+d×sum(incoming PR/outdegree).

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

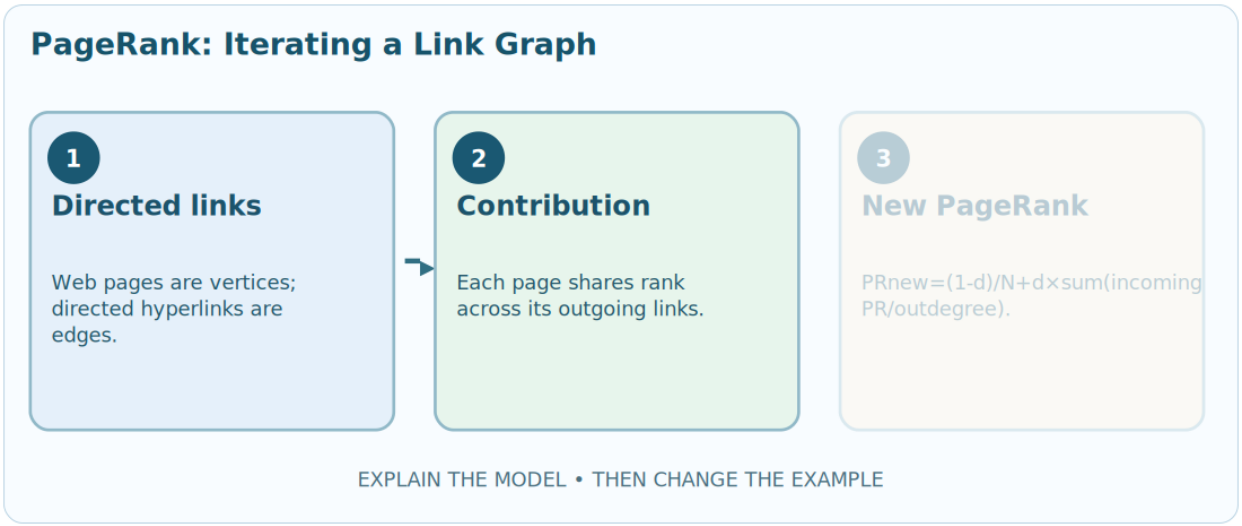
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



PageRank: Iterating a Link Graph: Directed links, Contribution, New PageRank

New Learning

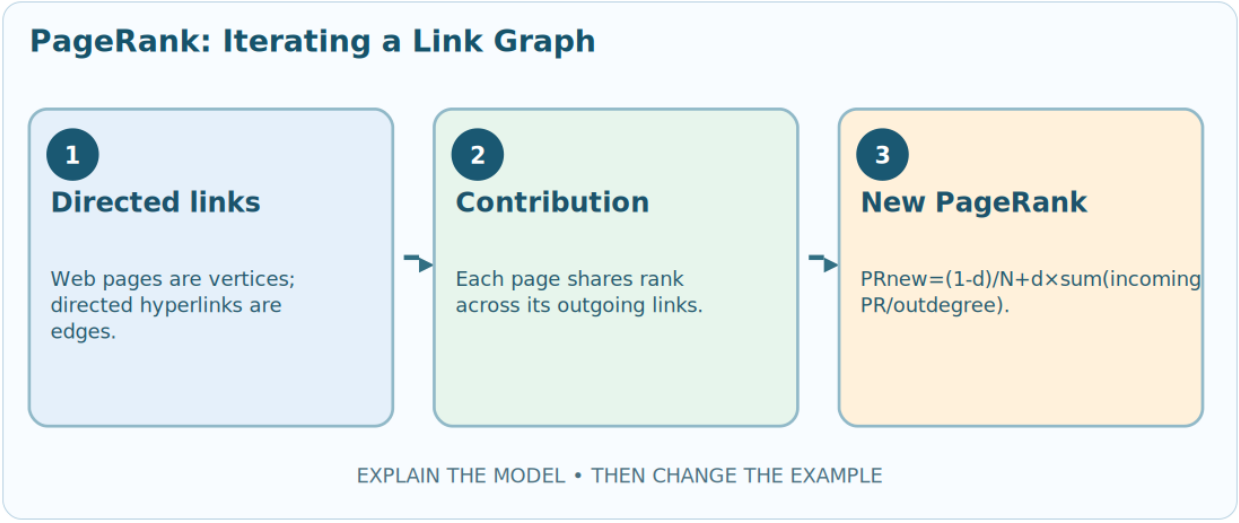


PageRank: Iterating a Link Graph: Directed links, Contribution, New PageRank

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



PageRank: Iterating a Link Graph: Directed links, Contribution, New PageRank

I DO — WORKED EXAMPLE

Follow the worked method

A→B,C; B→C; C→A. Initially PR(A)=PR(B)=PR(C)=1/3; d=0.85. Calculate one iteration.

A→B,C | B→C | C→A

Identify the input or diagram and predict the first result.

Incoming shares 1/3,1/6,1/2

Follow the second stage and explain its importance.

Result .333,.192,.475

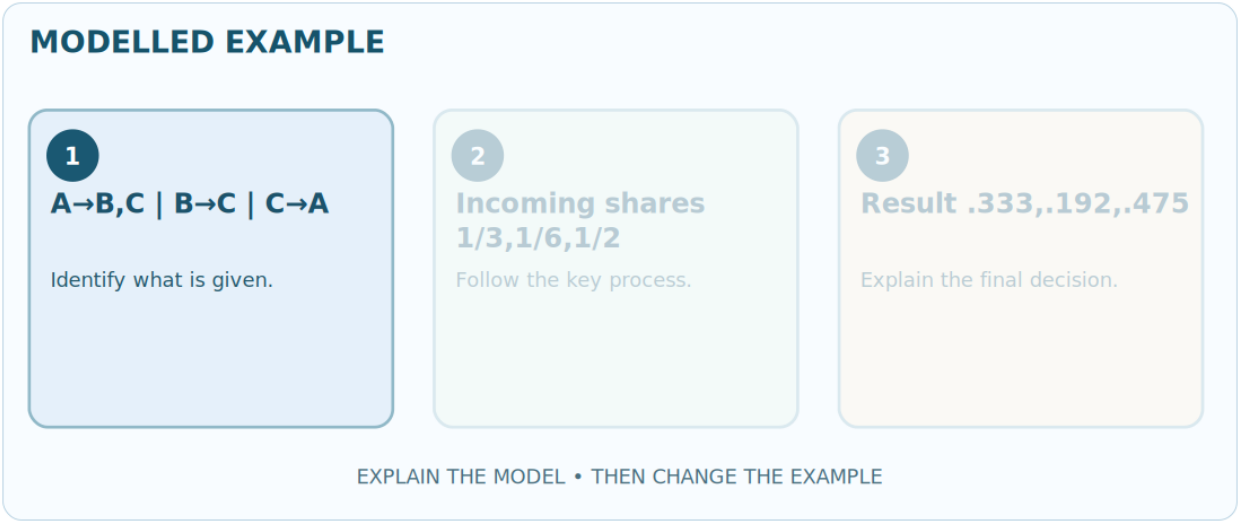
Incoming shares A:1/3; B:1/6; C:1/6+1/3=1/2. Base (1-.85)/3=.05. New A=.3333, B=.1917, C=.4750, sum=1.0000.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A→B,C; B→C; C→A. Initially PR(A)=PR(B)=PR(C)=1/3; d=0.85. Calculate one iteration.

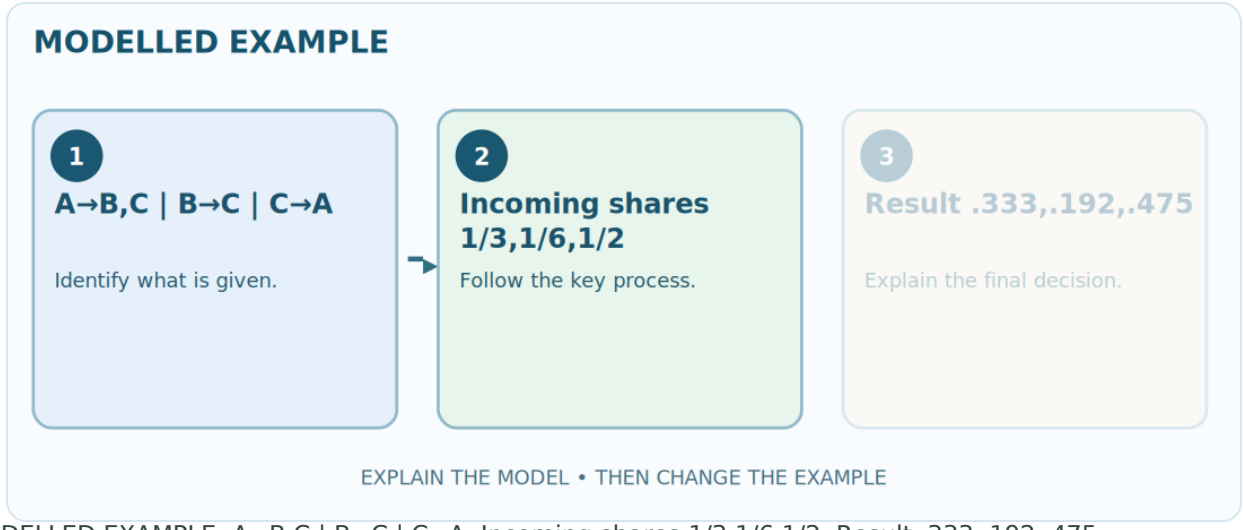
Identify the input or diagram and predict the first result.



MODELLED EXAMPLE: A→B,C | B→C | C→A, Incoming shares 1/3,1/6,1/2, Result .333,.192,.475

A→B,C; B→C; C→A. Initially PR(A)=PR(B)=PR(C)=1/3; d=0.85. Calculate one iteration.

Identify the input or diagram and predict the first result.



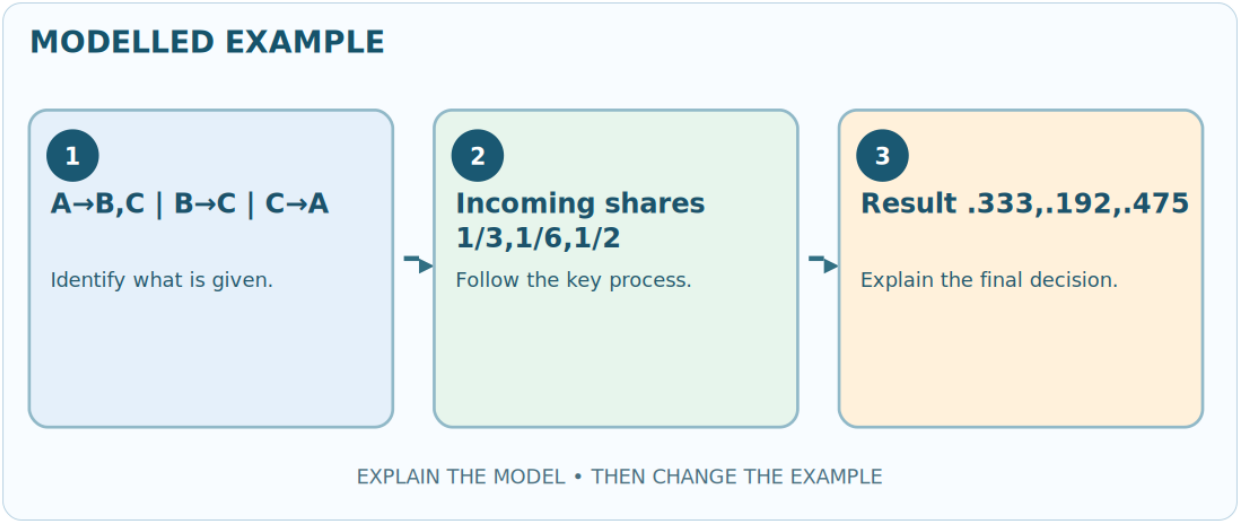
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I Do · original lesson diagrams

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A→B,C; B→C; C→A. Initially PR(A)=PR(B)=PR(C)=1/3; d=0.85. Calculate one iteration.

Identify the input or diagram and predict the first result.



MODELLED EXAMPLE: A→B,C | B→C | C→A, Incoming shares 1/3,1/6,1/2, Result .333,.192,.475

WE DO — GUIDED PRACTICE

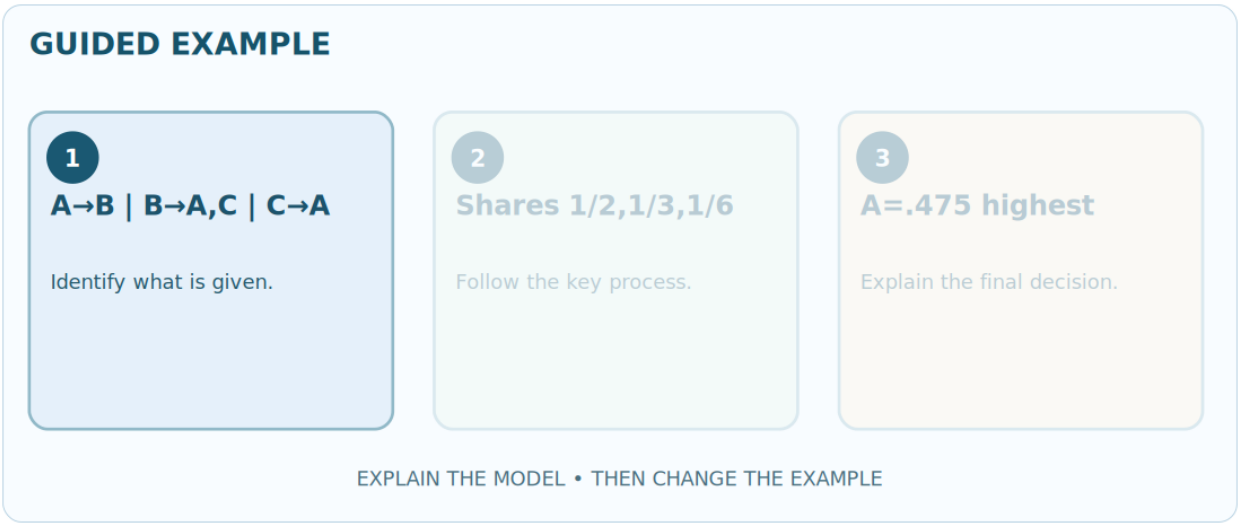
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A→B; B→A,C; C→A; equal initial rank and d=0.85. Which page ranks highest after one update?

Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: A→B | B→A,C | C→A, Shares 1/2,1/3,1/6, A=.475 highest

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: A→B | B→A,C | C→A Identify the input or diagram and predict the first result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain how PageRank uses inbound links and outbound link counts. [6]

Question 2

A page has rank 0.40 and two outgoing links. Calculate each undamped share, explain the division, and recalculate for four outgoing links. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compute one PageRank update for A→B,C B→C C→A with equal initial ranks and d=.85, then explain an important limitation. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

With PageRank a page with two outgoing links shares its current rank equally between them in the simplified model.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

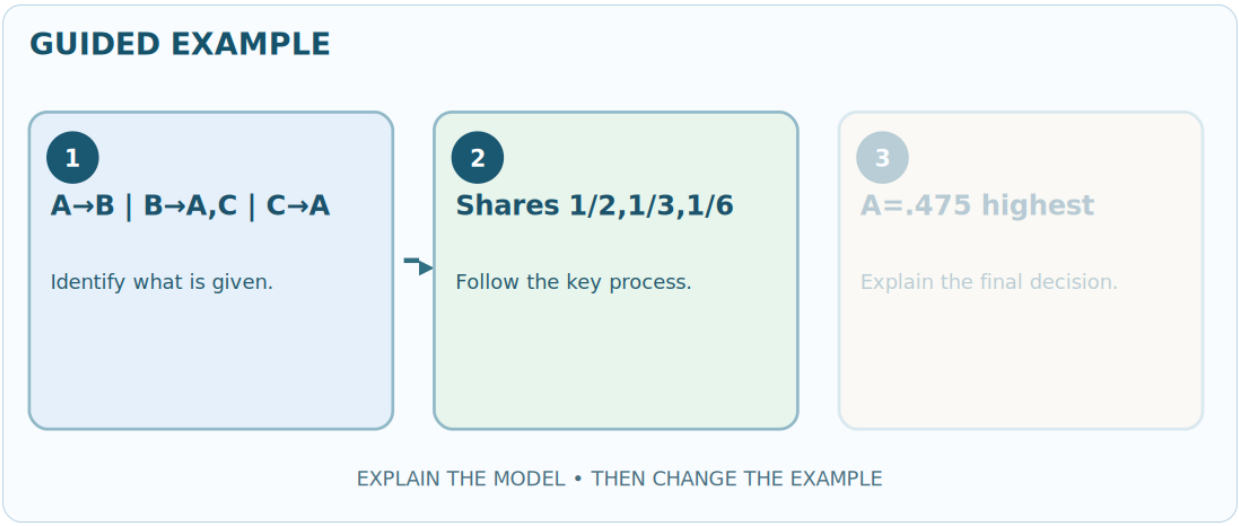
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A→B; B→A,C; C→A; equal initial rank and d=0.85. Which page ranks highest after one update?

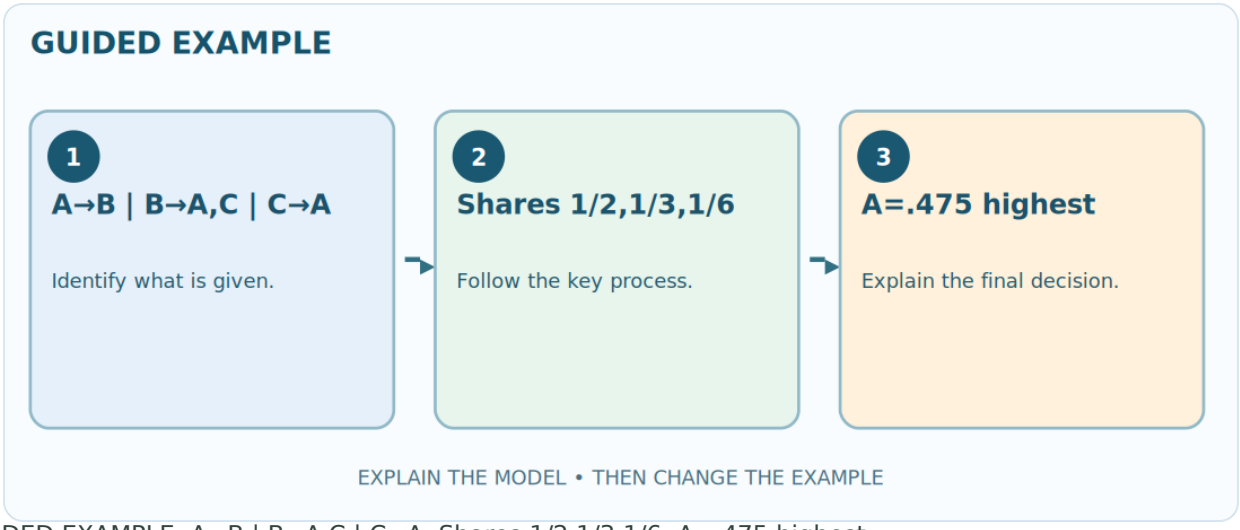
Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: A→B | B→A,C | C→A, Shares 1/2,1/3,1/6, A=.475 highest

A→B; B→A,C; C→A; equal initial rank and d=0.85. Which page ranks highest after one update?

Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: A→B | B→A,C | C→A, Shares 1/2,1/3,1/6, A=.475 highest

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Shares 1/2,1/3,1/6

Follow the second stage and explain its importance.

A=.475 highest

Incoming contributions A=1/6+1/3=1/2, B=1/3, C=1/6. Add base 0.05 and scale by 0.85: A=.4750, B=.3333, C=.1917. A ranks highest.

IMPROVE: Explain one correction to your own reasoning.